UNIVERSIDAD DE LAS FUERZAS ARMADAS – ESPE

Department of Computer Science
Software Engineering
36862 – Object-Oriented Programming

UNIT 1 PROJECT · TEAM WORK

Software Requirements Specification: ProvStock

Prov&Tec Management, a Comprehensive Sales and Inventory System

Team	ByteWorks
Members	Andrés Pachacama, Carlos Parra, Matias Paucar
NRC	36862
Instructor	Edison Lascano, PhD
Date	October 26, 2026
Academic period	SII-2026 (September 2026 – February 2027)
Version	1.3

Revision History

Table 1

Document Versions

Version	Date	Team	Author	Description of Changes
1.0	October 18, 2026	Team ByteWorks	Andrés Pachacama	Initial proposal of the requirements (FR-01 to FR-09).
1.1	October 19, 2026	Team ByteWorks	Carlos Parra	Structure aligned with IEEE Std 830-1998; inconsistencies between roles, requirements, and the use case diagram corrected; data requirements, business rules, role permissions, MoSCoW priorities, measurable non-functional requirements, and open issues added; APA 7 formatting applied.
1.2	October 20, 2026	Team ByteWorks	Matias Paucar	Problem statement consolidated (problems P1 to P7); FR-01.7, FR-02.6, FR-04.9, and FR-09.5 added; FR-03.4, FR-05.4, and FR-06.1 updated; product location field added (Table B2); role permissions (Table 3), use case diagram (Figure 1), and traceability matrix (Table C1) aligned with the 20 use cases; heading of Section 3.8 fixed; open issue OI-10 added.
1.3	October 26, 2026	Team ByteWorks	Andrés Pachacama	Requirements aligned with the implementation of ProvStock: the reports are exported to CSV instead of PDF (FR-09.4, OI-08); the data storage mechanism was defined as JSON files (Section 2.4, CON-02, DEP-04, SI-02); use case diagram (Figure 1) and traceability matrix (Table C1) updated with Export report to CSV, which is now included by Generate reports; acronyms CSV and JSON added (Table 2).

Table of Contents

Revision History 2
1. Introduction 5
1.1 Purpose 5
1.2 Intended Audience and Document Conventions 5
1.3 Product Scope 6
1.4 Definitions, Acronyms, and Abbreviations 7
1.5 Related Documents 7
2. Overall Description 8
2.1 Product Perspective 8
2.2 Product Functions 9
2.3 User Classes and Characteristics 9
2.4 Operating Environment 12
2.5 Design and Implementation Constraints 13
2.6 User Documentation 13
2.7 Assumptions and Dependencies 13
3. Functional Requirements 14
3.1 FR-01: Access Management 15
3.2 FR-02: Customer Management 16
3.3 FR-03: Product Management 17
3.4 FR-04: Sales Management 18
3.5 FR-05: Inventory Management 19
3.6 FR-06: Supplier and Purchase Management 20
3.7 FR-07: Sales History 21
3.8 FR-08: Stock Alerts 21
3.9 FR-09: Reports 22
3.10 Business Rules 23
4. Non-Functional Requirements 24
4.1 NFR-01: Performance 24
4.2 NFR-02: Security 24
4.3 NFR-03: Usability 25
4.4 NFR-04: Reliability and Data Integrity 25
4.5 NFR-05: Maintainability 25
4.6 NFR-06: Portability 26
4.7 NFR-07: Backup 26
5. External Interface Requirements 26
5.1 User Interfaces 26
5.2 Hardware Interfaces 27
5.3 Software Interfaces 27
5.4 Communication Interfaces 27
References 28

Appendix A 29
Appendix B 31
Appendix C 35
Appendix D 36

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements of ProvStock, the sales and inventory management system for Prov&Tec. The document describes what the system must do, who will use it, and the conditions under which it will operate. It is the reference for the client and the development team during the design, implementation, and testing of the system.

Version 1.3 is the requirements baseline used for the implementation of ProvStock; it updates version 1.2 with the decisions made during the implementation (JSON files to store the data and CSV files for the reports). The values marked as *(Proposed)* are decisions of the development team that the client, Ximena Añamise, has not yet confirmed explicitly; the pending decisions are listed in Appendix D.

1.2 Intended Audience and Document Conventions

This document is intended for three groups of readers: the client, who validates that the requirements reflect the needs of the business; the development team (Team ByteWorks), who use it to design, implement, and test the system; and the course instructor, who evaluates it. The client should read Sections 1, 2, and 3 and Appendix D; the development team should read the whole document.

The following conventions are used:

- **Identifiers.** Functional requirements are numbered FR-xx.y, where FR-xx is the feature and y is the individual requirement. Non-functional requirements use NFR-xx; business rules, BR-xx; interface requirements, UI-xx, HI-xx, and SI-xx; constraints, CON-xx; assumptions, AS-xx; dependencies, DEP-xx; and open issues, OI-xx. Each feature traces to one Product Backlog item (RF01 to RF09).
- **Requirement language.** “Shall” indicates a mandatory requirement and “should” indicates a desirable one, following ISO/IEC/IEEE 29148:2018 (International Organization for Standardization et al., 2018).
- **Priorities.** Priorities follow the MoSCoW scale: *Must have* (essential for the first release), *Should have* (important, but the system is usable without it), *Could have* (desirable if time allows), and *Won’t have* (excluded from this version).

- **Proposed values.** Items marked *(Proposed)* are suggestions of the development team that the client has not yet confirmed.
- **Format.** The document follows the formatting rules of the APA *Publication Manual* (American Psychological Association, 2020). Section numbers are kept, as recommended by IEEE Std 830-1998 (Institute of Electrical and Electronics Engineers [IEEE], 1998), so that requirements can be traced and referenced.

1.3 Product Scope

ProvStock is a desktop application that will organize and manage the sales and inventory operations of Prov&Tec, a company that sells electrical materials. The system will centralize the information about customers, products, suppliers, sales, and purchases; record where each product is kept in the warehouse; update the stock of each product every time a sale or a purchase is registered; warn the staff when a product reaches its minimum stock level; and produce sales, inventory, purchase, and slow-moving product reports.

The objectives of the system are the following:

- Replace the manual, paper-based management currently used by the company.
- Reduce the time customers wait to be served, which is currently between 10 and 20 minutes, to less than 2 minutes for a sale of up to five products (see NFR-03).
- Avoid purchasing materials that are already in stock by showing the real stock of each product at any time.
- Record which user performed each sale, purchase, and stock adjustment, so that every operation can be traced.
- Support the owner’s control of the business and decision-making through the sales history, alerts, and reports.

The following items are outside the scope of this version: online sales, payment processing, electronic invoicing with the Ecuadorian tax authority (SRI), accounting, management of multiple branches, a mobile application, and barcode scanning. Electronic invoicing is a growing trend in the market and a possible future extension, but it is not part of this version.

1.4 Definitions, Acronyms, and Abbreviations

Table 2 lists the acronyms and abbreviations used in this document. The business terms are defined in the glossary in Appendix A.

Table 2

Acronyms and Abbreviations

Term	Definition
BR	Business Rule. A rule of the business that the system must enforce.
CSV	Comma-Separated Values. A plain text format for tables, with one record per line; it can be opened with Excel. In ProvStock, the fields are separated by semicolons.
FR	Functional Requirement. A function the system must perform.
GUI	Graphical User Interface.
JRE	Java Runtime Environment. The software needed to run Java applications.
JSON	JavaScript Object Notation. A plain text format that stores data as objects with named fields; ProvStock uses it to save its information.
MoSCoW	Prioritization scale: Must have, Should have, Could have, Won't have.
MVC	Model–View–Controller. An architectural pattern that separates data, user interface, and control logic.
NFR	Non-Functional Requirement. A quality attribute or operational constraint of the system.
RF	Requisito Funcional (Spanish for "functional requirement"). Identifier of the Product Backlog items RF01 to RF09.
RUC	Registro Único de Contribuyentes. The 13-digit Ecuadorian taxpayer identification number.
SRI	Servicio de Rentas Internas. The Ecuadorian tax authority.
SRS	Software Requirements Specification.
VAT	Value-Added Tax (IVA, Impuesto al Valor Agregado, in Spanish).

1.5 Related Documents

This SRS is based on the project definition prepared in Workshop 07 (Pachacama et al., 2026b) and on the Product Backlog of ProvStock, which contains the requirements RF01 to RF09 (Pachacama et al., 2026a). Its structure follows IEEE Std 830-1998 (IEEE, 1998) and ISO/IEC/IEEE 29148:2018 (International Organization for Standardization et al., 2018). The complete references are listed in the References section.

2. Overall Description

2.1 Product Perspective

Prov&Tec currently manages its sales and inventory with paper documents, such as sales slips and order forms, and with basic spreadsheets. This model has allowed the company to sustain itself and grow, but it no longer supports the daily control of the business and causes the following problems, each of which is addressed by one or more features of the Product Backlog:

- **P1. Scattered information.** The data about customers, products, and suppliers is spread across papers and spreadsheets, with no single reliable source. Addressed by RF02, RF03, and RF06.
- **P2. Unknown real stock.** Incoming and outgoing goods are recorded late and inconsistently, and the location of the products in the warehouse is not recorded, so the staff do not know how many units of each product are available or where to find them. Addressed by RF03, RF04, RF05, and RF06.
- **P3. Unnecessary purchases.** Because the real stock is unknown, the company orders materials that it already has. Addressed by RF05, RF08, and RF09.
- **P4. Slow customer service.** Customers wait between 10 and 20 minutes while the salesperson searches for products, checks paperwork, and calculates the sale by hand, and many of them leave without buying. Addressed by RF02, RF03, and RF04.
- **P5. Uncontrolled restocking.** There is no timely warning of the products that are running out, which causes stockouts. Addressed by RF08.
- **P6. Decisions without data.** Slow-moving stock accumulates and generates losses, and the owner has no sales history or reports to know what is sold, what is bought, and how much the inventory is worth. Addressed by RF07 and RF09.
- **P7. No traceability.** Several people with different responsibilities handle the same information without access control, and there is no record of who registered a sale, received goods, or corrected the stock. Addressed by RF01, RF04, RF05, RF06, and RF07.

ProvStock is a new, self-contained system that replaces this manual process and solves the problems listed above; it is not part of a larger product. It brings together user access control; customer, product, and supplier management; sales and purchase registration; automatic stock

updates; the sales history; stock alerts; and reports in a single application. In this version, the system does not exchange information with external systems (see Section 5.4).

2.2 Product Functions

The main functions of ProvStock correspond to the nine features of the Product Backlog:

- **Access management (RF01):** log in and log out with a username and password, manage user accounts and roles, change passwords, and reset forgotten passwords.
- **Customer management (RF02):** register, update, search, view, and deactivate customers.
- **Product management (RF03):** register, update, search, and view the electrical products sold by the company, including their minimum stock level and their location in the warehouse.
- **Sales management (RF04):** register sales with their customer, products, and quantities; calculate the subtotal, VAT, and total; void sales registered by mistake; and update the VAT rate.
- **Inventory management (RF05):** update the stock automatically after each sale, purchase, or voided sale; record manual stock adjustments, including the load of the initial inventory; and show the current stock and the stock movements of each product.
- **Supplier and purchase management (RF06):** register, update, and deactivate suppliers, and register the purchases made to them.
- **Sales history (RF07):** view and filter all registered sales with their date, customer, and salesperson.
- **Stock alerts (RF08):** identify the products whose stock is at or below the minimum level and show an alert.
- **Reports (RF09):** generate sales, inventory, purchase, and slow-moving product reports for a selected period.

2.3 User Classes and Characteristics

The system has three user roles. Their permissions are summarized in Table 3, and their interaction with the system is shown in Figure 1.

Salesperson

The salesperson serves customers at the store counter and is the most frequent user of the system. This role registers sales, searches for products and their stock, and registers or updates customer information. Salespeople have basic computer skills, and they need to complete each sale quickly.

Warehouse Clerk

The warehouse clerk is in charge of the warehouse. This role registers and updates products, records their location in the warehouse, registers the goods received from suppliers, adjusts the stock when there are differences, reviews the stock movements, and receives the alerts about products with low stock.

Owner

The owner is responsible for the business as a whole and can perform all the functions of the salesperson and the warehouse clerk. In addition, the owner manages user accounts (including resetting forgotten passwords), manages suppliers, deactivates customers, voids sales, updates the VAT rate, reviews the sales history, and generates reports to make decisions. The owner, Ximena Añamise, is also the client who validates this document.

In Figure 1, the abstract actor User represents any person with an account; the functions shared by all roles (logging in, logging out, changing the password, and searching for products) are assigned to it. Besides these roles, the system itself checks the stock levels automatically after every stock change, without a user starting the process (see FR-08).

Table 3

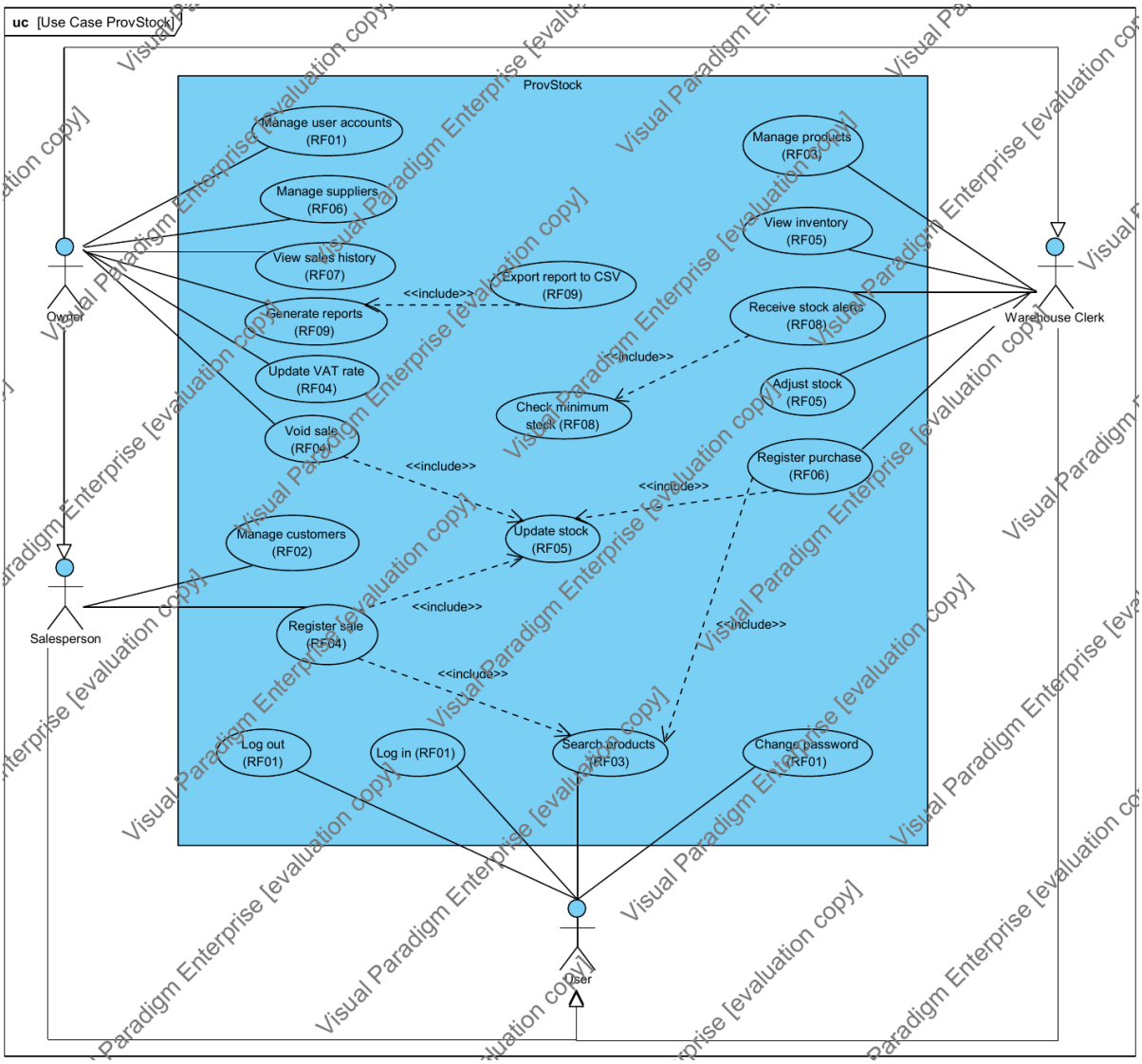
Role Permissions (Proposed)

Function	Salesperson	Warehouse Clerk	Owner
Log in, log out, and change own password	Yes	Yes	Yes
Manage user accounts and reset passwords	—	—	Yes
Register and update customers	Yes	—	Yes
Deactivate customers	—	—	Yes
Search and view products and stock	Yes	Yes	Yes
Register, update, and deactivate products	—	Yes	Yes
Register sales	Yes	—	Yes
Void sales	—	—	Yes
Update the VAT rate	—	—	Yes
Adjust stock manually	—	Yes	Yes
View inventory and stock movements	—	Yes	Yes
Register, update, and deactivate suppliers	—	—	Yes
Register purchases	—	Yes	Yes
View sales history	—	—	Yes
Receive stock alerts	—	Yes	Yes
Generate reports	—	—	Yes

Note. Yes = the role can perform the function; — = the function is not available to the role.

Figure 1

Use Case Diagram of ProvStock



Note. The abstract actor User represents any person with an account. The hollow arrows indicate actor generalization: the Salesperson and the Warehouse Clerk specialize User (with one shared triangle), and the Owner specializes both of them, so the Owner inherits all their use cases. «include» marks a use case that is always performed as part of another one; The codes in parentheses refer to the Product Backlog items.

2.4 Operating Environment

ProvStock will be a desktop application developed in Java. It will run on the computers available at the store, using a Java Runtime Environment, on Windows 10 or later (Proposed). The system stores users, customers, products, suppliers, sales, purchases, and stock movements

in JSON files, one file per type of data, in the data folder of the application; the files are read and written with the Gson library, which is included in the application. The reports are saved as CSV files in the reports folder.

The client has not yet confirmed whether the system will be used on a single computer or on several computers (for example, one at the counter and another in the warehouse). In the second case, all computers must share the same data through the local network of the store. This decision is pending (OI-03 in Appendix D).

2.5 Design and Implementation Constraints

CON-01 The system shall be developed in Java, applying the object-oriented principles taught in the course (encapsulation, inheritance, polymorphism, and abstraction).

CON-02 The data shall be stored in JSON files, a technology covered in the course, so that no database management system has to be installed.

CON-03 The system shall be developed with free or open-source tools, so the client does not have to pay for licenses *(Proposed)*.

CON-04 The system shall be completed within the academic term by a team of three developers; therefore, the scope is limited to the nine Product Backlog features.

2.6 User Documentation

The system shall be delivered with a short user manual in Spanish that explains the main tasks of each role, and the interface shall show clear messages that guide the user when an error occurs *(Proposed)*.

2.7 Assumptions and Dependencies

The following assumptions are considered for the proposed system:

AS-01 Users will enter accurate information when registering customers, products, suppliers, sales, and purchases.

AS-02 Users have basic computer skills and will receive a short training session before using the system.

AS-03 Sales and purchases will be registered when they happen, so the stock stays up to date.

AS-04 The owner or the warehouse clerk will define a minimum stock level for each product.

AS-05 Before the system goes into operation, the staff will count the physical inventory and load the initial stock of each product through a stock adjustment (FR-05.4).

AS-06 The store has at least one computer that meets the requirements described in Sections 2.4 and 5.2.

The operation of the system depends on the following conditions:

DEP-01 Sales registration depends on the customers and products previously registered in the system (FR-02 and FR-03).

DEP-02 Stock alerts depend on the minimum stock level registered for each product (FR-03 and FR-08).

DEP-03 The sales history and the reports depend on the sales, purchases, and stock movements previously recorded (FR-04 to FR-06).

DEP-04 The application depends on a compatible Java Runtime Environment and on the Gson library, which is distributed with the application.

DEP-05 The final version of the requirements depends on the availability of the client to review and validate this document (Appendix D).

3. Functional Requirements

This section describes the functional requirements of ProvStock, grouped by the nine features of the Product Backlog. For each feature, it states the description, the actors, the priority, the Product Backlog reference, the individual requirements, and the acceptance criteria. The data that the system stores for each entity are detailed in Appendix B, and the business rules that apply to all features are listed in Section 3.10.

3.1 FR-01: Access Management

Description. The system shall control who can use it and which functions each user can access according to their role.

Actors. Salesperson, Warehouse Clerk, and Owner (the Owner also manages user accounts and resets passwords).

Priority. Must have.

Product Backlog reference. RF01.

Requirements.

- **FR-01.1** The system shall support three user roles, Salesperson, Warehouse Clerk, and Owner, with the permissions defined in Table 3.
- **FR-01.2** The system shall require each user to log in with a username and password before accessing any function.
- **FR-01.3** The system shall reject a login attempt with an incorrect username or password and show a general error message that does not reveal which of the two was wrong.
- **FR-01.4** The system shall allow the owner to create, update, and deactivate user accounts and to assign a role to each account.
- **FR-01.5** The system shall allow each user to change their own password by entering the current password and the new password twice.
- **FR-01.6** The system shall allow users to log out.
- **FR-01.7** The system shall allow the owner to reset the password of a user account by entering a new temporary password of at least 8 characters, without knowing the previous password *(Proposed)*.

Acceptance criteria.

- A user with valid credentials logs in and sees only the functions of their role.
- A login attempt with an incorrect username or password is rejected.
- A deactivated user cannot log in.
- After a password change, only the new password is accepted.
- After the owner resets a user's password, the user can log in only with the new password.

3.2 FR-02: Customer Management

Description. The system shall allow users to manage the information of the customers who buy from the company.

Actors. Salesperson and Owner (only the Owner can deactivate customers).

Priority. Must have.

Product Backlog reference. RF02.

Requirements.

- **FR-02.1** The system shall allow users to register a new customer with the data defined in Table B1.
- **FR-02.2** The system shall verify that the required fields are complete and that the identification number is unique and has a valid length (10 digits for a cédula or 13 digits for a RUC).
- **FR-02.3** The system shall allow users to update the data of an existing customer.
- **FR-02.4** The system shall allow users to search for customers by identification number or name and view their information.
- **FR-02.5** The system shall provide a default “Final Consumer” customer for sales in which the buyer does not provide personal data *(Proposed)*.
- **FR-02.6** The system shall allow the owner to deactivate a customer; a deactivated customer cannot be selected in new sales but remains in the history. The “Final Consumer” customer cannot be modified or deactivated.

Acceptance criteria.

- A customer registered with valid data appears in the customer list.
- Registering a customer with an identification number that already exists is rejected with a clear message.
- The changes made to a customer are shown the next time the customer is viewed.
- Searching by identification number or by part of the name returns the matching customers.
- A deactivated customer cannot be selected when a new sale is registered.

3.3 FR-03: Product Management

Description. The system shall allow users to manage the electrical products sold by the company.

Actors. Warehouse Clerk and Owner (register and update); all roles (search and view).

Priority. Must have.

Product Backlog reference. RF03.

Requirements.

FR-03.1 The system shall allow users to register a new product with the data defined in Table B2, including its minimum stock level.

FR-03.2 The system shall verify that the product code is unique, that the sale price is greater than zero, and that the purchase cost and the minimum stock level are not negative.

FR-03.3 The system shall allow users to update the information of a product, except its current stock, which only changes through the operations described in FR-05.

FR-03.4 The system shall allow users to search for products by code, name, or category and show their sale price, current stock, and location in the warehouse.

FR-03.5 The system shall allow users to deactivate a product that is no longer sold; a deactivated product cannot be added to new sales or purchases, but it remains in the history.

Acceptance criteria.

- A product registered with valid data appears in the product list with its stock and minimum stock level.
- Registering a product with a code that already exists is rejected.
- Searching by code, name, or category returns the matching products with their price, stock, and location.
- A deactivated product does not appear when a new sale is registered.

3.4 FR-04: Sales Management

Description. The system shall allow users to register the sales made to customers.

Actors. Salesperson and Owner (only the Owner can void a sale or update the VAT rate).

Priority. Must have (FR-04.8 and FR-04.9: Should have).

Product Backlog reference. RF04.

Requirements.

FR-04.1 The system shall allow users to register a new sale and shall automatically record its date, time, and the logged-in salesperson.

FR-04.2 The system shall require the user to select the customer of the sale, either a registered customer or the "Final Consumer", and shall allow registering a new customer without leaving the sale.

FR-04.3 The system shall allow the user to add products to the sale, indicating the quantity of each one, and shall show the unit price and the available stock of each product.

FR-04.4 The system shall not allow a quantity greater than the available stock of the product (BR-03).

FR-04.5 The system shall calculate the subtotal of each line, the subtotal of the sale, the VAT, and the total amount, according to BR-05 and BR-06.

FR-04.6 The system shall record the payment method of the sale (cash, bank transfer, or card) for information purposes only *(Proposed)*.

FR-04.7 The system shall assign a unique consecutive number to each registered sale.

FR-04.8 The system shall allow the owner to void a registered sale by entering a reason; the voided sale shall remain in the history with the status "Voided", and its stock shall be restored (FR-05.3).

FR-04.9 The system shall allow the owner to update the VAT rate; the new rate shall apply only to the sales registered after the change, and each sale shall keep the rate that was applied to it (BR-05).

Acceptance criteria.

- A salesperson registers a sale with a customer and at least one product, and the sale is saved with its number, date, and salesperson.

- For a sale with a subtotal of $10.00, the system shows a VAT of $1.50 and a total of $11.50.
- Entering a quantity greater than the available stock is rejected with a message that shows the available stock.
- After a sale is voided, it appears as “Voided” in the history and the stock of its products returns to the previous value.
- After the owner changes the VAT rate, new sales use the new rate, and the sales registered before the change keep their original rate.

3.5 FR-05: Inventory Management

Description. The system shall keep the stock of each product up to date and allow its control.
Actors. Warehouse Clerk and Owner; the system updates the stock automatically.
Priority. Must have.
Product Backlog reference. RF05.
Requirements.

FR-05.1 The system shall decrease the stock of each product by the quantity sold when a sale is registered.

FR-05.2 The system shall increase the stock of each product by the quantity received when a purchase is registered.

FR-05.3 The system shall restore the stock of the products of a sale when the sale is voided.

FR-05.4 The system shall allow the warehouse clerk and the owner to adjust the stock of a product manually (for example, to load the initial inventory before the system goes into operation, or because of damaged or lost items or differences found in a physical count), entering the quantity and the reason; the system shall record the user and the date of the adjustment.

FR-05.5 The system shall show the current stock of all products and highlight the products whose stock is at or below the minimum level.

FR-05.6 The system shall keep a record of every stock movement (sale, purchase, voided sale, or adjustment) for each product.

Acceptance criteria.

- After a sale of 3 units, the stock of the product decreases by 3; after a purchase of 10 units, it increases by 10.
- A manual adjustment cannot be saved without a reason.
- The stock shown for a product equals its initial stock plus purchases, minus sales, plus voided sales, plus or minus adjustments.

3.6 FR-06: Supplier and Purchase Management

Description. The system shall allow users to manage suppliers and register the purchases made to them.

Actors. Owner (suppliers); Warehouse Clerk and Owner (purchases).

Priority. Must have.

Product Backlog reference. RF06.

Requirements.

FR-06.1 The system shall allow the owner to register, update, and deactivate suppliers with the data defined in Table B3, verifying that the RUC is unique; a deactivated supplier cannot be selected in new purchases.

FR-06.2 The system shall allow users to search for suppliers by RUC or name and view their information.

FR-06.3 The system shall allow users to register a purchase indicating the supplier, the date, the supplier's invoice number (optional), and the products with their quantities and unit costs; the system shall record the user who registered it.

FR-06.4 The system shall calculate the total cost of each purchase.

FR-06.5 The system shall assign a unique consecutive number to each registered purchase.

Acceptance criteria.

- The owner registers a supplier with valid data, and the supplier appears in the supplier list.
- A purchase with at least one product is saved with its number, supplier, date, and total cost, and the stock of its products increases.
- A purchase without a supplier or without products cannot be saved.
- A deactivated supplier cannot be selected when a new purchase is registered.

3.7 FR-07: Sales History

Description. The system shall allow the owner to review all the registered sales.

Actors. Owner.

Priority. Should have.

Product Backlog reference. RF07.

Requirements.

- **FR-07.1** The system shall list all registered sales, showing their number, date, customer, salesperson, total amount, and status.
- **FR-07.2** The system shall allow the owner to filter the sales by date range, salesperson, customer, or status.
- **FR-07.3** The system shall show the details of a selected sale: its products, quantities, unit prices, subtotal, VAT, and total.

Acceptance criteria.

- The owner sees every registered sale, including voided ones, with its date and responsible salesperson.
- Filtering by a date range shows only the sales made within that range.
- Opening a sale shows the same products and amounts that were registered.

3.8 FR-08: Stock Alerts

Description. The system shall monitor the stock levels and alert the users when a product needs to be restocked.

Actors. System (automatic check); Warehouse Clerk and Owner (receive the alerts).

Priority. Should have.

Product Backlog reference. RF08.

Requirements.

- **FR-08.1** The system shall check whether the stock of a product is at or below its minimum stock level after every stock change (sale, purchase, voided sale, or adjustment) and every time a user logs in.

FR-08.2 The system shall show an alert to the warehouse clerk and the owner for each product at or below the minimum level, indicating its current stock and minimum stock level.

FR-08.3 The system shall show the number of active alerts in the main menu and allow the user to open the list of alerts.

FR-08.4 The system shall remove the alert of a product automatically when its stock rises above the minimum level.

Acceptance criteria.

- When a sale leaves a product with a stock equal to or lower than its minimum level, an alert for that product is shown to the warehouse clerk and the owner.
- The salesperson does not receive stock alerts.
- After a purchase raises the stock above the minimum level, the alert disappears.

3.9 FR-09: Reports

Description. The system shall generate reports that summarize the operations of the business.

Actors. Owner.

Priority. Should have (FR-09.4 and FR-09.5: Could have).

Product Backlog reference. RF09.

Requirements.

FR-09.1 The system shall generate a sales report for a selected date range, showing the number of sales, the total sold, the VAT collected, the sales per salesperson, and the best-selling products.

FR-09.2 The system shall generate an inventory report showing, for each product, its current stock, its minimum stock level, and the value of the stock at purchase cost.

FR-09.3 The system shall generate a purchase report for a selected date range, showing the purchases per supplier and their total cost.

FR-09.4 The system shall save every generated report as a CSV file that can be opened with Excel *(Proposed)*.

FR-09.5 The system shall generate a slow-moving products report for a selected date range, listing the active products that had no sales in that period, with their current stock and the value of that stock at purchase cost *(Proposed)*.

Acceptance criteria.

- The totals of the sales report match the sum of the non-voided sales in the selected range.
- The inventory report shows the same stock as the inventory screen.
- The purchase report shows only the purchases made within the selected range.
- The slow-moving products report lists only the active products without sales in the selected range.
- Every generated report is saved as a CSV file in the reports folder, with the same data shown on the screen and with the date and time in the file name.

3.10 Business Rules

The following rules apply to all the features described above:

BR-01 Each sale must have exactly one customer (registered or "Final Consumer"), one salesperson, and at least one product with a quantity greater than zero.

BR-02 Each purchase must refer to one registered supplier and include at least one product with a quantity and a unit cost greater than zero.

BR-03 A sale cannot include a quantity of a product greater than its available stock; therefore, the stock can never be negative.

BR-04 Registered sales and purchases cannot be edited or deleted. A sale registered by mistake can only be voided by the owner (FR-04.8), and an error in a purchase is corrected through a stock adjustment (FR-05.4).

BR-05 The VAT rate is 15%, the general rate in force in Ecuador in 2026 (Sánchez, 2025). Because this rate can be changed by executive decree, the owner shall be able to update it, and each sale shall keep the rate that was applied to it.

BR-06 Prices and costs are expressed in U.S. dollars with two decimals and are registered without VAT. The VAT is calculated on the subtotal of the sale and rounded to two decimals *(Proposed)*.

BR-07 Customers, products, suppliers, and users that appear in registered sales or purchases cannot be deleted; they can only be deactivated.

BR-08 Usernames, customer identification numbers, product codes, and supplier RUCs must be unique.

4. Non-Functional Requirements

This section describes the quality requirements of ProvStock. Each requirement includes measurable acceptance criteria so it can be verified during testing.

4.1 NFR-01: Performance

The system shall complete basic operations (logging in, registering, searching, and updating records) within 3 seconds, and shall generate reports within 10 seconds, with a volume of data of at least 1,000 products, 500 customers, and 10,000 sales.

Priority. Must have.

Acceptance criteria.

- Basic operations are completed within 3 seconds, and reports within 10 seconds, in tests with the indicated volume of sample data.

4.2 NFR-02: Security

The system shall protect user accounts and enforce the role permissions. Passwords shall have at least 8 characters, shall be hidden while they are typed, and shall be stored as salted hashes (never in plain text). A user shall not be able to open a function that is not allowed for their role, even if the function exists in the application.

Priority. Must have.

Acceptance criteria.

- The stored data do not contain any password in plain text.
- A password with fewer than 8 characters is rejected.
- The functions that are not allowed for a role are neither shown nor executable for that role.

4.3 NFR-03: Usability

The user interface shall be in Spanish, the language of the users, and shall be simple and consistent. A trained salesperson shall be able to register a sale of up to five products in less than 2 minutes, which addresses the current waiting time of 10 to 20 minutes. Error messages shall explain what is wrong and how to correct it.

Priority. Must have.

Acceptance criteria.

- In a test with a trained user, a sale of five products is registered in less than 2 minutes.
- After a training session of one hour, a new user can complete the main tasks of their role without help.
- All menus, buttons, and messages are in Spanish.

4.4 NFR-04: Reliability and Data Integrity

The system shall save each operation as soon as the user confirms it and shall keep the information when the application is closed and reopened. The registration of a sale, a voided sale, or a purchase shall be atomic: either the operation and all its stock changes are saved, or none of them are.

Priority. Must have.

Acceptance criteria.

- The records saved before closing the application are available after restarting it.
- If an error occurs while a sale is being saved, neither the sale nor any stock change is stored.
- The stock of every product matches the formula described in the acceptance criteria of FR-05.

4.5 NFR-05: Maintainability

The system shall use an object-oriented design with separate classes for its main entities (for example, User, Role, Customer, Supplier, Product, Sale, SaleDetail, Purchase, PurchaseDetail, and StockMovement) and shall separate the user interface, the business logic, and the data access, for example with the MVC pattern. The source code shall follow the Java naming conventions.

Priority. Should have.

Acceptance criteria.

- Each main entity is represented by its own class with clearly defined responsibilities.
- The data storage mechanism can be changed by modifying only the data-access classes.

4.6 NFR-06: Portability

The system shall run on Windows 10 or later with a compatible Java Runtime Environment *(Proposed)* and shall not require an Internet connection.

Priority. Should have.

Acceptance criteria.

- The application is installed and runs correctly on a computer with Windows 10 or later and no Internet connection.

4.7 NFR-07: Backup

The system should allow the owner to create a backup copy of the data and to restore it *(Proposed)*.

Priority. Could have.

Acceptance criteria.

- A backup created by the owner can be restored, and the restored data match the data at the moment of the backup.

5. External Interface Requirements

This section describes how users interact with ProvStock and how the system communicates with the hardware and software it needs.

5.1 User Interfaces

UI-01 The system shall provide a login window that asks for the username and password.

UI-02 The system shall provide a main menu that shows only the functions allowed for the role of the logged-in user, together with the number of active stock alerts.

UI-03 The system shall provide forms to register and update customers, products, suppliers, users, sales, and purchases; the required fields shall be marked, and the validation messages shall appear next to the field with the error.

UI-04 The system shall show the registered information in lists or tables that can be searched and sorted.

UI-05 The sales screen shall allow the user to search for products, add them to the sale, and see the subtotal, VAT, and total update immediately, all in a single window.

UI-06 The system shall show messages to confirm completed operations, report errors, and show the stock alerts.

UI-07 The interface shall use Spanish, dates in the format dd/mm/yyyy, and amounts in U.S. dollars with two decimals.

5.2 Hardware Interfaces

HI-01 The system shall work with a standard keyboard and mouse.

HI-02 The system shall be displayed correctly on a monitor with a resolution of at least 1366 × 768 pixels.

HI-03 The system shall use the storage device of the computer (or of the computer that hosts the shared database) to save the information.

5.3 Software Interfaces

SI-01 The system shall run on the Java Runtime Environment and on the operating system described in NFR-06.

SI-02 The system shall save and retrieve its information through a data-access layer connected to the selected storage mechanism (JSON files), so the mechanism can be replaced without changing the rest of the system.

5.4 Communication Interfaces

This version does not require integration with external services, such as online sales, online payments, or electronic invoicing with the SRI, and it does not need an Internet connection. If the client decides to use more than one computer (OI-03), the computers shall communicate with the shared database through the local network of the store.

References

American Psychological Association. (2020). *Publication manual of the American Psychological Association* (7th ed.). https://doi.org/10.1037/0000165-000

Institute of Electrical and Electronics Engineers. (1998). *IEEE recommended practice for software requirements specifications* (IEEE Std 830-1998). https://doi.org/10.1109/IEEESTD.1998.88286

International Organization for Standardization, International Electrotechnical Commission, & Institute of Electrical and Electronics Engineers. (2018). *Systems and software engineering—Life cycle processes—Requirements engineering* (ISO/IEC/IEEE Standard No. 29148:2018). https://doi.org/10.1109/IEEESTD.2018.8559686

Pachacama, A., Parra, C., & Paucar, M. (2026a). *Product backlog: ProvStock* [Unpublished manuscript]. Department of Computer Science, Universidad de las Fuerzas Armadas ESPE.

Pachacama, A., Parra, C., & Paucar, M. (2026b). *Workshop 07—Project definition: Prov&Tec management: Comprehensive sales and inventory system* [Unpublished manuscript]. Department of Computer Science, Universidad de las Fuerzas Armadas ESPE.

Sánchez, E. (2025, December 29). *IVA se mantiene en 15% en Ecuador para 2026 si el Ejecutivo no emite un decreto* [VAT remains at 15% in Ecuador for 2026 if the Executive does not issue a decree]. El Diario. https://www.eldiario.ec/centro/iva-ecuador-15-por-ciento-2026-20251229/

Appendix A

Glossary

Table A1 defines the business terms used in this document.

Table A1

Glossary of Terms

Term	Definition
Alert	A notification that a product has reached or fallen below its minimum stock level.
Cédula	The 10-digit Ecuadorian national identity card number.
Customer	A person or organization that buys products from the company.
Final Consumer	A generic customer record used for sales in which the buyer does not provide personal data (consumidor final).
Inventory	The set of products and the stock of each one available in the warehouse.
Location	The place in the warehouse where a product is kept, for example, an aisle and a shelf.
Minimum stock level	The lowest quantity of a product that should remain in the warehouse before it has to be restocked.
Owner	The user responsible for the business, who has access to all functions of the system.
Product	An electrical material sold by the company and registered in the system.
Prov&Tec	The company that sells electrical materials and will use the system.
Purchase	A record of the products bought from a supplier, with their quantities and costs.
Report	A summary of sales, inventory, or purchase information for a selected period.
Sale	A record of the products sold to a customer by a salesperson on a given date.
Sales history	The list of all the sales registered in the system.
Salesperson	A user who serves customers and registers sales.
Slow-moving product	An active product that has had no sales during a selected period.
Stock	The quantity of a product currently available.
Stock adjustment	A manual change to the stock of a product, with a reason, to correct differences with the physical inventory.
Stock movement	Any change in the stock of a product caused by a sale, a purchase, a voided sale, or an adjustment.
Supplier	A person or company from which the business buys products.

Term	Definition
Voided sale	A registered sale cancelled by the owner; it remains in the history, and its stock is restored.
Warehouse clerk	A user in charge of the products and the stock in the warehouse.

Appendix B

Data Requirements (Proposed)

Tables B1 to B7 describe the data that the system shall store for each entity. These fields are a proposal of the development team and must be confirmed with the client (OI-01).

Table B1

Customer Data

Field	Type	Required	Rules and Notes
Identification type	List	Yes	Cédula, RUC, or passport.
Identification number	Text	Yes	Unique; 10 digits for a cédula, 13 for a RUC.
Full name or business name	Text	Yes	
Phone number	Text	No	
Email	Text	No	Must have a valid email format.
Address	Text	No	
Status	List	Yes	Active or inactive; active by default.

Table B2

Product Data

Field	Type	Required	Rules and Notes
Product code	Text	Yes	Unique.
Name	Text	Yes	For example, “Cable sólido #12 AWG”.
Description	Text	No	
Category	List	Yes	For example, cables, lighting, switches and outlets, protection devices, or tools.
Brand	Text	No	
Location	Text	No	Place in the warehouse where the product is kept; for example, “Aisle 3 – Shelf B”.
Unit of measure	List	Yes	Unit, meter, roll, or box.
Purchase cost	Decimal	Yes	U.S. dollars without VAT; zero or greater.
Sale price	Decimal	Yes	U.S. dollars without VAT; greater than zero.
Current stock	Decimal	Yes	Changed only through stock movements (FR-05); never negative.
Minimum stock level	Decimal	Yes	Zero or greater.
Status	List	Yes	Active or inactive.

Note. Stock values are decimals because some products, such as cables, are sold by the meter.

Table B3

Supplier Data

Field	Type	Required	Rules and Notes
RUC	Text	Yes	Unique; 13 digits.
Business name	Text	Yes	
Contact person	Text	No	
Phone number	Text	Yes	
Email	Text	No	Must have a valid email format.
Address	Text	No	
Status	List	Yes	Active or inactive.

Table B4

Sale Data

Field	Type	Required	Rules and Notes
Sale number	Number	Auto	Unique and consecutive.
Date and time	Date	Auto	Taken from the system clock.
Salesperson	Reference	Auto	The logged-in user.
Customer	Reference	Yes	Registered customer or “Final Consumer”.
Lines	List	Yes	At least one; each line has a product, quantity, unit price, and line subtotal.
Subtotal, VAT rate, VAT, and total	Decimal	Auto	Calculated according to BR-05 and BR-06.
Payment method	List	Yes	Cash, bank transfer, or card.
Status	List	Auto	Registered or voided.
Void reason, date, and user	Text	If voided	Entered by the owner when the sale is voided.

Table B5

Purchase Data

Field	Type	Required	Rules and Notes
Purchase number	Number	Auto	Unique and consecutive.
Date	Date	Yes	Today by default.
Registered by	Reference	Auto	The logged-in user.
Supplier	Reference	Yes	An active registered supplier.
Supplier invoice number	Text	No	
Lines	List	Yes	At least one; each line has a product, quantity, unit cost, and line subtotal.
Total cost	Decimal	Auto	Sum of the line subtotals.

Table B6

User Account Data

Field	Type	Required	Rules and Notes
Username	Text	Yes	Unique.
Full name	Text	Yes	
Role	List	Yes	Salesperson, warehouse clerk, or owner.
Password	Text	Yes	At least 8 characters; stored as a salted hash (NFR-02).
Status	List	Yes	Active or inactive.

Table B7

Stock Movement Data

Field	Type	Required	Rules and Notes
Date and time	Date	Auto	Taken from the system clock.
Product	Reference	Yes	
Type	List	Auto	Sale, purchase, voided sale, or adjustment.
Quantity	Decimal	Yes	Positive for entries and negative for exits.
Reference document	Reference	Auto	The related sale or purchase, if any.
User	Reference	Auto	The logged-in user.
Reason	Text	Adjustments	Required for manual adjustments (FR-05.4).

Appendix C

Requirements Traceability Matrix

Table C1 relates each feature of this document to the Product Backlog item it comes from, the use cases of Figure 1, the roles involved, and its priority.

Table C1

Requirements Traceability Matrix

Feature	Backlog	Use Cases (Figure 1)	Roles	Priority
FR-01 Access Management	RF01	Log in; Log out; Change password; Manage user accounts	All; Owner	Must
FR-02 Customer Management	RF02	Manage customers	Salesperson, Owner	Must
FR-03 Product Management	RF03	Search products; Manage products	All; Warehouse Clerk, Owner	Must
FR-04 Sales Management	RF04	Register sale; Void sale; Update VAT rate	Salesperson, Owner; Owner; Owner	Must
FR-05 Inventory Management	RF05	View inventory; Adjust stock; Update stock	Warehouse Clerk, Owner; System	Must
FR-06 Supplier and Purchase Management	RF06	Manage suppliers; Register purchase	Owner; Warehouse Clerk, Owner	Must
FR-07 Sales History	RF07	View sales history	Owner	Should
FR-08 Stock Alerts	RF08	Check minimum stock; Receive stock alerts	System; Warehouse Clerk, Owner	Should
FR-09 Reports	RF09	Generate reports; Export report to CSV	Owner	Should

Appendix D

Validation and Open Issues

The requirements in this document constitute Version 1.3 of the proposal for ProvStock. They will be reviewed with the client, Ximena Añamise, to confirm that the system addresses the sales and inventory management needs of Prov&Tec. The requested changes will be evaluated and incorporated into later versions of this SRS, and the requirements will be considered final once the client approves them.

Table D1 lists the questions that must be answered by the client during the review.

Table D1

Open Issues to Validate with the Client

ID	Question	Affects
OI-01	Are the customer, product, and supplier fields proposed in Appendix B the ones the business needs?	FR-02, FR-03, FR-06
OI-02	Are prices registered with or without VAT, and do all products use the 15% rate?	FR-04, BR-05, BR-06
OI-03	Will the system be used on one computer or on several computers (for example, at the counter and in the warehouse)?	Section 2.4, Section 5.4
OI-04	Can sales be registered for a “Final Consumer” without personal data?	FR-02.5, BR-01
OI-05	Which payment methods should be recorded?	FR-04.6
OI-06	Who can void a sale, and is there a time limit to do it?	FR-04.8
OI-07	Do the proposed MoSCoW priorities agree with the priorities of the Product Backlog?	Section 3
OI-08	Which information should each report include, and is the CSV export enough?	FR-09
OI-09	Who defines the minimum stock level of each product?	FR-03, FR-08
OI-10	Should a user be required to change a password reset by the owner the next time they log in?	FR-01.7

Client approval

Name: Ximena Añamise, Owner of Prov&Tec

Signature: _____

Date: _____